

SUZ-AK36NLHZ



3/8" / 5/8"

HP-1

-38 FT

Attic

SVZ-AP36NL 8.0 ft

0.0 ft

37,429 BTU/h (28,472 BTU/h)

Est Cooling Discharge Air Temp: 48.6 °
Est Heating Discharge Air Temp: 107.1 °

Refrigerant

System Refrigerant:

R454B

Additional Field -Charge:

0.00 lb

Total System Charge:

9.92 lb

Min. Applied Floor Area:

149.7 ft²

Max System Charge:

9.92 lb

Correction Factors

Cooling Heating

Outdoor Temperature:

1.05 0.94

Piping Length:

1.00 1.00

Defrosting:

- 1.00

User Derate:

1.00 1.00

Outdoor Altitude:

1.00 1.00

1.04 0.93

Conditions

Cooling

Indoor Dry Bulb

78.0 °F

Indoor Wet Bulb

66.0 °F

Indoor Humidity

53.5 %

Heating

Indoor Dry Bulb

72.0 °F

Indoor Wet Bulb

3.0 °F

Indoor Humidity

3.0 %

Outdoor Dry Bulb

89.0 °F

Outdoor Humidity

100.0 %

HP-1 MAIN HOUSE

MUZ-FX12NLHZ



1/4" / 3/8"

HPOU-2

12.0ft (3)

MSZ-FX12NL

Annex HPIU-2

12,493 BTU/h (9,274 BTU/h)

Est Cooling Discharge Air Temp: 59.7 °
Est Heating Discharge Air Temp: 100.6 °

System Refrigerant:

R454B

Additional Field -Charge:

0.00 lb

Total System Charge:

2.76 lb

Min. Applied Floor Area:

49.9 ft²

Max System Charge:

3.30 lb

Correction Factors

Cooling Heating

Outdoor Temperature:

1.05 1.12

Piping Length:

1.00 1.00

Defrosting:

- 1.00

User Derate:

1.00 1.00

Outdoor Altitude:

1.00 1.00

1.04 1.11

Conditions

Cooling

Indoor Dry Bulb

78.0 °F

Indoor Wet Bulb

66.0 °F

Indoor Humidity

53.5 %

Heating

Indoor Dry Bulb

72.0 °F

Indoor Wet Bulb

3.0 °F

Indoor Humidity

3.0 %

Outdoor Dry Bulb

89.0 °F

Outdoor Humidity

100.0 %

HP-2 ANNEX

HEAT PUMP SYSTEM SCHEMATICS

NTS

HEATING COOLING AND VENTILATION LOADS				
AREA (SF)	MAIN HVAC UNIT	HEATING LOAD (MBH)	TOTAL COOLING LOAD (MBH / TONS)	NOTES
MAIN HOUSE FIRST FLOOR	HPIU-1	20.9	13.6 / 1.1	INCLUDES VENTILATION
BASEMENT	NONE	15.2		INCLUDES VENTILATION
ANNEX	HPIU-2	12.9	0.6 / 0.5	INCLUDES VENTILATION
MAIN HOUSE ATTIC	HPIU-1	2.2		ASSUMES R-50 ROOF AND END WALLS. ATTICS WITHIN CONDITIONED SPACE

OUTSIDE DESIGN CONDITIONS			
SUMMER	WINTER	NOTES	
92°F DBT / 71°F WBT	6°F DBT	FRAMINGHAM, MA	

INSIDE DESIGN CONDITIONS			
AREAS	WINTER	SUMMER	NOTES
FIRST FLOOR, ANNEX	72 °F	74 °F	BASEMENT, ATTIC CONDITIONED TO 55 °F, 80 °F

- DUCT UP THROUGH FLOOR
- DUCT DOWN THROUGH FLOOR
- SUPPLY DIFFUSER
- RETURN/EXHAUST REGISTER
- MANUAL VOLUME DAMPER
- MOTORIZED DAMPER
- WIRELESS THERMOSTAT
- WIRELESS ERV RETURN BOOST SWITCH
- RETURN AIR FLOW DIRECTION
- SUPPLY AIR FLOW DIRECTION
- EXISTING/HIDDEN DUCTWORK
- NEW DUCTWORK & DIFFUSERS

HVAC LEGEND

- CAP CAPACITY
- CD CEILING DIFFUSER
- CFM CUBIC FEET PER MINUTE - AIR FLOW RATE
- COP COEFFICIENT OF PERFORMANCE
- D DEHUMIDSTAT
- DBT DRY BULB TEMPERATURE
- EAO HRV - EXHAUST AIR OUT
- EFFEC HRV EFFECTIVENESS
- EG EXHAUST GRILLE
- FPM FEET PER MINUTE
- FPIU HEAT PUMP INDOOR UNIT
- HPDU HEAT PUMP OUTDOOR UNIT
- HRV HEAT RECOVERY VENTILATOR
- HSPF2 (IV) HEATING SEASON PERFORMANCE FACTOR, VERSION 2, ASHRAE REGION 4
- IN WG INCHES OF WATER, GAUGE PRESSURE
- KW KILOWATTS
- KBH THOUSAND BTUS PER HOUR
- MBH THOUSAND BTUS PER HOUR
- Ø DIAMETER
- OAI HRV - OUTSIDE AIR IN
- OAT OUTSIDE AIR TEMPERATURE
- RAI HRV - RETURN AIR IN
- RAP RETURN AIR PATH DEVICE - SEE SCHEDULE
- RG RETURN GRILLE - SEE SCHEDULE
- RHC REHEAT COIL
- RR RETURN REGISTER - SEE SCHEDULE
- SAO ERV - SUPPLY AIR OUT
- SD SUPPLY DIFFUSER CEILING - SEE SCHEDULE
- SEER SEASONAL ENERGY EFFICIENCY RATIO
- SG SUPPLY GRILLE WALL - SEE SCHEDULE
- SR SUPPLY REGISTER WALL - SEE SCHEDULE
- T THERMOSTAT
- TV75 TERMINAL VELOCITY 75 FPM
- TYP TYPICAL
- V VENTILATION - ERV SUPPLY OR RETURN
- Vb VENTILATION BOOST SWITCH
- Vc VENTILATION CO2 CONTROL
- VD VOLUME DAMPER
- W WATTS
- WBT WET BULB TEMPERATURE

HVAC SYMBOLS

THE DRAWINGS ARE DIAGRAMMATIC AND INDICATE GENERALLY THE LOCATION OF MATERIALS AND EQUIPMENT. THE CONTRACTOR SHALL FURNISH AND INSTALL ALL LABOR, EQUIPMENT, AND MATERIALS SHOWN AND REQUIRED TO HAVE A COMPLETE AND FUNCTIONING SYSTEM TOGETHER WITH ALL ASSOCIATED APPARATUS AS SHOWN ON THE PLANS. "PROVIDE" SHALL MEAN TO FURNISH AND INSTALL.

INFORMATION AND COMPONENTS SHOWN ON FLOW DIAGRAMS AND DETAIL DRAWINGS BUT NOT SHOWN ON FLOOR PLANS, AND VICE VERSA, SHALL APPLY AND SHALL BE PROVIDED AS THOUGH REQUIRED EXPRESSLY BY BOTH. IT IS NOT INTENDED TO SHOW EVERY OFFSET, FITTING OR COMPONENT; HOWEVER, CONTRACT DOCUMENTS REQUIRE COMPONENTS AND MATERIALS WHETHER OR NOT INDICATED OR SPECIFIED AS NECESSARY TO MAKE THE INSTALLATION COMPLETE AND OPERATIONAL.

FINAL LOCATION OF EQUIPMENT, AND CONNECTION POINTS SHALL BE DETERMINED IN THE FIELD WITH THE CONTRACTOR BEING RESPONSIBLE FOR DIMENSIONS THAT SHALL BE CONFIRMED AND CORRELATED AT THE JOB SITE.

THE WORK SHALL BE DONE IN ACCORDANCE WITH THE LOCAL BUILDING MECHANICAL CODES, NFPA AND ALL OTHER APPLICABLE STATE AND FEDERAL CODES.

INSTALL WORK SO AS TO BE READILY ACCESSIBLE FOR OPERATION, MAINTENANCE, AND REPAIR. MINOR DEVIATIONS FROM THE DRAWINGS MAY BE MADE TO ACCOMPLISH THESE GOALS.

ALL DUCT AND TERMINALS SHALL BE INSTALLED APPROXIMATELY AS INDICATED AND MODIFIED TO SUIT BUILDING CONDITIONS TO AVOID INTERFERENCE, TO MAINTAIN CLEARANCE AND MINIMIZE PRESSURE DROP. DUCT FITTINGS AT THE AHU DUCT DISCHARGE AND RETURN PLENUMS AND END OF DUCT DISTRIBUTION ARE INDICATED TO MINIMIZE PRESSURE DROP.

GUARANTEE WORK IN WRITING FOR ONE YEAR FROM DATE OF FINAL NOTICE OF ACCEPTANCE. REPAIR OR REPLACE DEFECTIVE MATERIALS, EQUIPMENT, WORKMANSHIP AND INSTALLATION THAT DEVELOP WITHIN THIS PERIOD, PROMPTLY AND TO THE OWNERS SATISFACTION AND CORRECT DAMAGE CAUSED IN MAKING NECESSARY REPAIRS AND REPLACEMENTS UNDER GUARANTEE WITHIN CONTRACT.

ASSURE ADEQUATE SUPPORT FOR DUCT AND EQUIPMENT. PREVENT VIBRATION AND SWAYING.

PREVENT ENTRY OF FOREIGN MATTER DURING CONSTRUCTION.

GENERAL NOTES

Owners Vander Straeten, Ghosh Wayland, MA	
HVAC Design OCTOBER ENGINEERING LLC Sudbury, MA 01776 508-561-7553	
ENERGY RETROFIT AND ELECTRIFICATION 13 Rice Spring Lane Wayland, MA 01778	
HVAC SCHEMATICS NOTES	
10-10-2025	H-1 1 of 5